

USAYPT 2026 KICKOFF

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AGENDA

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What is USAYPT

- Team-based physics research competition
- You investigate open-ended problems (not plug-and-chug)
- Present + defend like actual scientists

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COMPETITION FORMAT

PHYSICS FIGHTS



Tournament competition consists of “physics fights” in which presenters are questioned by an opposing team.

TEAM ROLES



Teams for each question will consist of one or more presenters and multiple scouts

Question 1

Friction On a Roll

In basic physics, friction is often idealized as either static or kinetic, and dependent on only a few variables. Real tires exhibit more complex behavior. Explore experimentally and theoretically how rolling and braking friction¹ depend on tire inflation pressure, speed, and surface conditions. Develop a model describing energy losses due to deformation and slip, and analyze how the conditions that minimize rolling friction differ from those that maximize braking performance. Extrapolate your work to automobile tires, and discuss the corresponding implications for fuel efficiency and carbon emissions.



QUESTION 2

Standing Waves

Explore how the frequency of standing waves on a string depends on the length, tension, linear density, and amplitude. Study an open string experimentally, here meaning a string under tension between two mechanically fixed end points (such as the bridge and nut of a guitar). Then consider a string whose effective vibrating length is shortened, and tension increased, by pressing it against the neck of an instrument with fixed frets (again a guitar provides a common example). Does the increase in tension have a measurable effect on the frequency? Measure the frequencies of open and fretted strings across a musical instrument, and compare them to the intended notes of the tuning system (for example, equal temperament). Consider whether the location of the frets should be adjusted.

QUESTION 3

Problem 3

Diodes

Diodes can be made from crystals such as galena, oxidized metals, or other semiconducting materials. Construct some, investigate their current-voltage characteristics, and study their reproducibility. Develop a model that explains their observed rectifying behavior, and identify the main sources of variability. Compare your results with the performance of modern commercial diodes. Discuss the applications of different types of diode. Consider using your home-made devices in a “foxhole radio.”

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Question 4

Diodes

Diodes can be made from crystals such as galena, oxidized metals, or other semiconducting materials. Construct some, investigate their current-voltage characteristics, and study their reproducibility. Develop a model that explains their observed rectifying behavior, and identify the main sources of variability. Compare your results with the performance of modern commercial diodes. Discuss the applications of different types of diode. Consider using your home-made devices in a “foxhole radio.”